Ssohrab Borhanian, Ph.D.

sborhanian@gmail.com | sborhanian.github.io | | EU Citizen

EXECUTIVE SUMMARY

- Open-source, scientific software developer with background in Python and ML tools.
- Physicist with 10+ years of academic expertise in gravitational-wave astrophysics and large-scale collaborations.
- Avid collaborator and communicator with strong academic publication and presentation record.
- Problem-solver and keen learner at heart; highly adaptable to new environments and challenges.

EDUCATION

The Pennsylvania State University *2014 – 2021*

Ph.D. and M.Sc. in Physics · Fulbright Scholar · Peter Clay Eklund Lecturing Award *State College, PA, USA*

Technical University of Berlin *2009 – 2014*

B.Sc. in Physics and Mathematics · Only double degree in my cohort *Berlin, Germany*

OPEN SOURCE PROJECTS

gwbench — Gravitational-Wave Detector and Network Benchmarking *2019 – Present*

Lead Developer and Maintainer · · Class. Quant. Grav. 38, 2021, [arXiv:2010.15202]

- Open-source, Python package for Fisher-Matrix-based benchmarking of gravitational-wave detector networks.
- Adopted as a primary analysis tool by the Einstein Telescope and Cosmic Explorer collaborations.
- Highly modular and streamlined user experience for large-scale computational studies.

xkn — Kilonova Modeling Framework *2021 – 2023*

Core Developer · · Mon. Not. Roy. Astron. Soc. 529 (2024) 1, 647-663,, [arXiv:2311.15709]

- Python package to generate kilonova light curves from binary neutron star mergers.
- *Key contributions*: Implementation of accessible user interface and optimization for computational efficiency.

EXPERIENCE

Postdoctoral Researcher *09/2023 – Present*

Università degli Studi di Milano-Bicocca · Research group: Prof. Davide Gerosa *Milan, Italy*

- Developing JAX-based ML-pipeline for gravitational-wave data analysis.
- Extending the Python package PRECESSION to support `jit`-compilation and GPU-acceleration through JAX.
- Investigated the impact of facility timing on the science case of third-generation, gravitational-wave detectors.

Postdoctoral Researcher *06/2021 – 08/2023*

Friedrich-Schiller-Universität Jena · Research group: Prof. Sebastiano Bernuzzi *Jena, Germany*

- Contributed to and extended the functionality of the open-source Python packages GWBENCH, XKN, and BAJES.

Graduate Researcher (Ph.D.) *08/2016 – 05/2021*

The Pennsylvania State University · Advisor: Prof. Bangalore S. Sathyaprakash *State College, PA, USA*

- Developed GWBENCH as a modular Python framework for large-scale science case studies.
- Investigated the science case of third-generation, gravitational-wave detectors via comprehensive trade studies.
- Investigated numerical relativity waveforms for a post-Newtonian signature in the strong-field regime.

Research Intern *08/2012 – 09/2012*

Duke University · RISE Worldwide Scholarship · Advisor: Prof. Kate Scholberg *Durham, NC, USA*

TECHNICAL SKILLS

Programming	Python, Swift, SwiftUI, Fortran, Java, Mathematica, MATLAB.
Python Ecosystem	astropy, h5py, jax, joblib, matplotlib, mpmath, numpy, pandas, pytest, scipy, sympy.
Developer Tools	Git/GitHub/GitLab, HPC clusters (SLURM/HTCondor), LaTeX.
Data Analysis	Machine Learning, Bayesian inference, Monte Carlo methods, time series analysis.

SOFT SKILLS

Communication	20+ presentations and invited seminars at international conferences and universities.
Technical Writing	25 academic publications (3700+ citations).
Languages	*Fluent:* German, English, Persian; *Basic:* French, Italian.